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Div:- D Roll no.:-

Course/Program: BTech C.S.E SY

Subject: Digital Logic and Microprocessor.

Practical - 01

* Aim:

To verify and interpret the logic and truth table for OR gate, AND gate, NOT gate, NAND gate, NOR gate, EX-OR gate and EX-NOR gate.

* Theory:-

- 1) AND gate: The AND gate is an electronic circuit that gives a high output (1) only if all its inputs are high. A dot (.) is used to show the AND operation i.e. $A \cdot B$ or can be written as AB

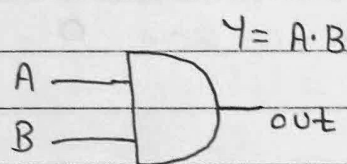


fig: Logic Symbol
of AND Gate.

I/P		O/P
A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

fig: Truth Table.

- 2) OR gate: The OR gate is an electronic circuit that gives a high output (1) if one or more of its inputs are high. A plus (+) is used to show the OR operation.

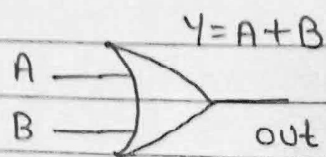


fig: Logic Symbol of OR gate.

Truth Table:

I/P		O/P
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

- 3) NOT gate: The NOT gate is an electronic circuit that produces an inverted version of the input at its output. It is also known as an inverter. If the input variable is A, the inverted output is known as NOT A. This is also shown as A' or $\bar{A}$, as shown at the outputs.

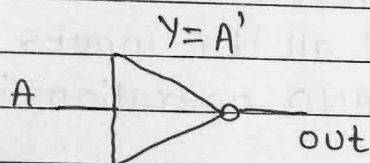


fig: Logic Symbol of NOT gate

Truth Table:

I/P		O/P
A	$\bar{A}$	$Y = \bar{A}$
0	1	1
1	0	0

- 4) NAND gate: This is a NOT-AND gate which is equal to an AND gate followed by a NOT gate. The output of all NAND gates are high if any of the inputs are low. The symbol is an AND gate with a small circle on the output. The small circle represents inversion.

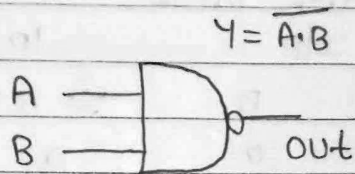


fig: Logic symbol of
NAND gate.

Truth Table:

I/P		O/P
A	B	$Y = \overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

- 5) NOR gate: This is a NOT-OR gate which is equal to an OR gate followed by a NOT gate. The outputs of all NOR gates are low if any of the inputs are high. The symbol is an OR gate with a small circle on the output. The small circle represents inversion.

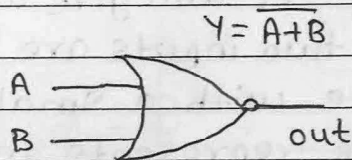


fig: Logic Symbol of
NOR gate

Truth Table:

I/P		O/P
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

- 6) EX-OR gate: The 'exclusive-OR' gate is a circuit which will give a high output if either, but not both of its two inputs are high. An encircled plus sign ($\oplus$) is used to show the EX-OR operation.

$$Y = A \oplus B$$



fig: Logic symbol of
Ex-OR gate.

Truth Table:

I/P		O/P
A	B	$Y = A \text{ XOR } B$
0	0	0
0	1	1
1	0	1
1	1	0

Ex-OR gate is created from AND, NAND and OR gates. The output is high only when both the inputs are different.

7) Ex-NOR gate: The 'Exclusive-NOR' gate circuit does the opposite to the Ex-OR gate. It will give a low output if either, but not of its two inputs are high. The symbol is an Ex-OR gate with a small circle on the output. The small circle represents invertible.

$$Y = \overline{A \oplus B}$$

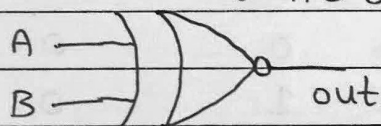


fig: Logic Symbol of
EX-NOR gate.

Truth Table:

I/P		O/P
A	B	$Y = \overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

Ex-NOR gate is created from AND, NOT and OR gates. The output is high only when both the inputs are same.

* Conclusion: We have successfully gone through the logic gates and their truth tables. We have studied the Pin diagram of the ICs. By understanding all the ICs, we have implemented them.

1. Pin diagram of AND Gate.

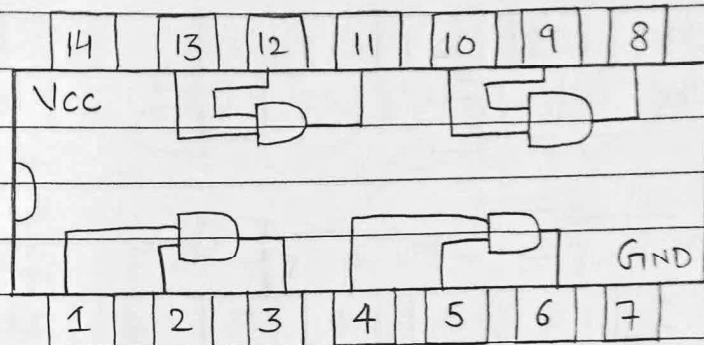


Fig: 7408 Quad 2 input AND Gates.

2. Pin diagram of OR Gate.

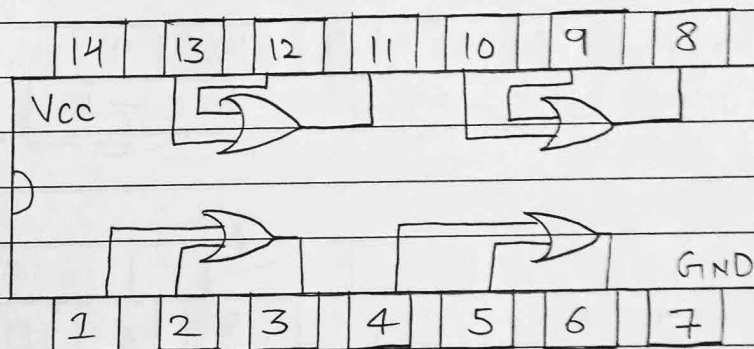


Fig: 7432 Quad 2 input OR Gates.

3. Pin diagram of NOT gate.

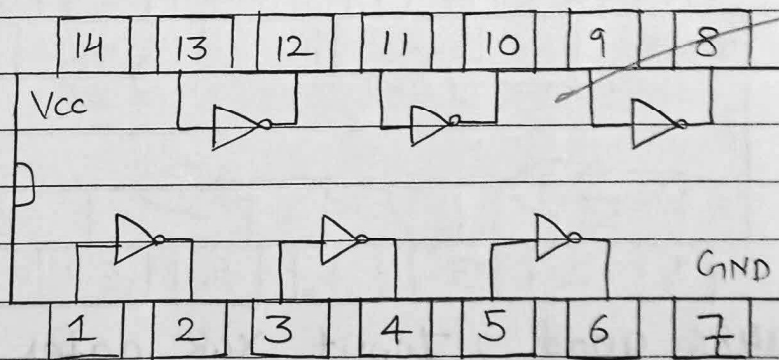


Fig: 7404 Six Inverter.

4. Pin diagram of NAND gate.

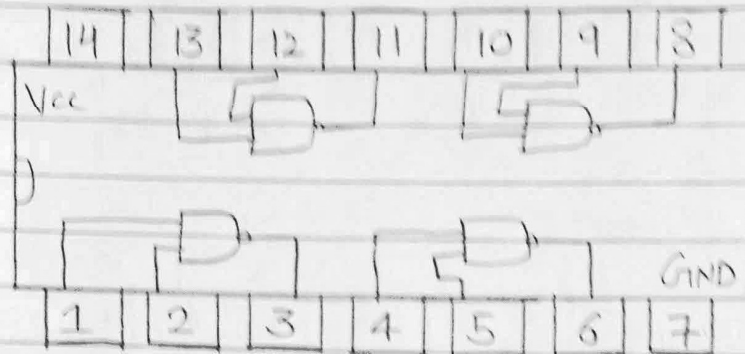


Fig: 7400 Quad 2 input NAND Gates.

5. Pin diagram of NOR gate.

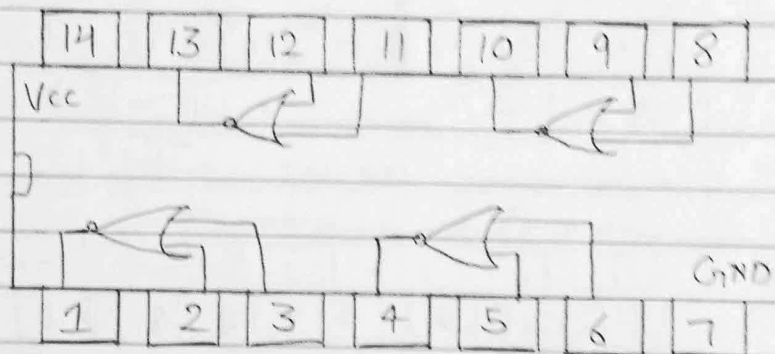


Fig: 7402 Quad 2 Input NOR Gates.

6. Pin diagram of XOR gate.

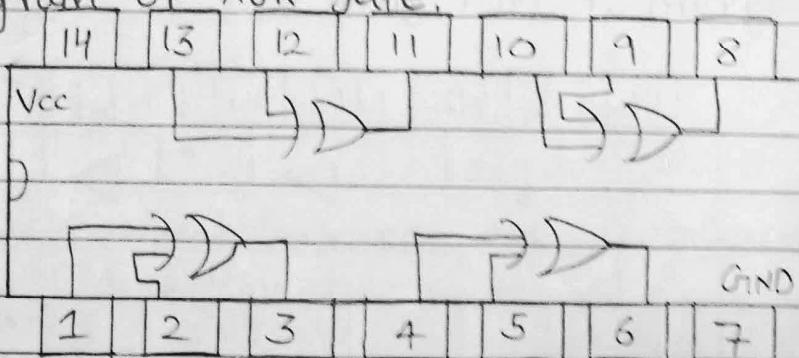


Fig: 7486 Quad 2 Input XOR gates.



7. Pin diagram of XNOR gate.

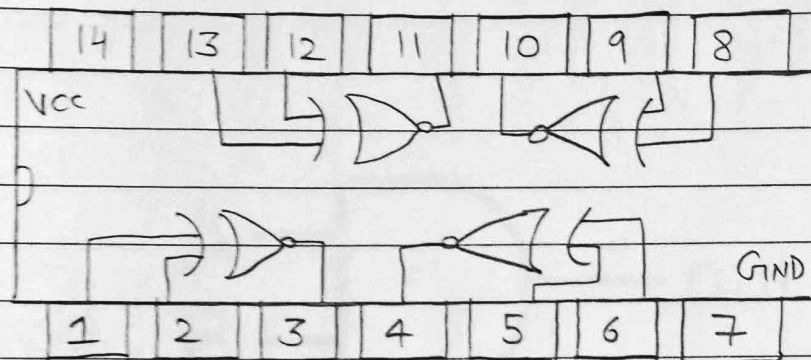


Fig: 747266 Quad 2 Input XNOR Gates.

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